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PARTICLES CHARACTERIZATION IN FLOW CYTOMETRY

Abstract

A method of characterizing particles in flow cytometry includes determining a first pulse width value of a particle using a first technique. The method includes determining a second pulse width value of the particle using a second technique. The method further includes comparing the first and second pulse width values, and characterizing the particle as a concatenated particle when a difference between the first and second pulse width values exceeds a threshold.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation of U.S. patent application Ser. No. 18/880,001, filed on Dec. 30, 2024, which is a U.S. National Stage Application of PCT International Application No. PCT/US2024/027828, filed on May 3, 2024, which claims the benefit of and priority to U.S. Provisional Application No. 63/501,943, filed on May 12, 2023; the disclosures of which hereby incorporated by reference in their entireties. To the extent appropriate, a claim of priority is made to each of the above-disclosed applications.

BACKGROUND

[0002] In flow cytometry, particles are arranged in a sample stream to pass in a single file line through one or more excitation light beams. Light that is scattered and/or emitted by the particles from interaction with the one or more excitation light beams is collected and analyzed to characterize and differentiate the particles. In a sorting flow cytometer, particles may be extracted out of the sample stream after having been characterized by their interaction with the one or more excitation beams, and thereby sorted into different groups.

[0003] Doublets occur when two cells pass through the one or more excitation light beams at the same time. Typically, doublets and other types of concatenated cells (e.g., triplets, quadruplets, and the like) are excluded from analysis in flow cytometry because they affect the quality of data such as by causing false positives and/or false negatives in the data.

SUMMARY

[0004] In general terms, the present disclosure relates to characterizing particles in flow cytometry. In one possible configuration, the particles are characterized to distinguish singlets from other types of particles based on a difference between first and second pulse width values that are calculated using different techniques. Various aspects are described in this disclosure, which include, but are not limited to, the following aspects.

[0005] One aspect relates to a method of characterizing particles in flow cytometry, the method comprising: determining a first pulse width value of a particle using a first technique; determining a second pulse width value of the particle using a second technique; comparing the first and second pulse width values; and characterizing the particle as a concatenated particle when a difference between the first and second pulse width values exceeds a threshold.

[0006] Another aspect relates to system for characterizing particles, the system comprising: a light emitting unit generating one or more excitation light beams; a fluidic system streaming the particles through the one or more excitation light beams at an interrogation zone; optical elements directing scattered light from the interrogation zone; one or more detectors measuring characteristics of the scattered light; and a processing circuitry having non-transitory computer readable storage media storing instructions which, when executed by the processing circuitry, cause the processing circuitry to: determine first pulse width values of the particles using a first technique; determine second pulse width values of the particles using a second technique; compare the first and second pulse width values; and characterize the particles as concatenated particles when a difference between the first and second pulse width values exceeds a threshold.

[0007] Another aspect relates to a method of characterizing a particle in flow cytometry, the method comprising: obtaining a waveform of the particle; determining a mean value of a baseline on both sides of the waveform of the particle; subtracting the mean value from data points of the waveform; determining a maximum value of the waveform; calculating a half maximum value by dividing the maximum value by two; subtracting the half maximum value from the data points of the waveform; obtaining absolute values for the data points to generate a folded waveform; determining a first minimum and a last minimum in the folded waveform; generating a first vector by performing linear regression on a first set of data points in proximity to the first minimum;

generating a second vector by performing linear regression on a second set of data points in proximity to the last minimum; and measuring a direct pulse width value of the particle as a distance between the first and second vectors at the half maximum value.

[0008] A variety of additional aspects will be set forth in the description that follows. The aspects can relate to individual features and to combination of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

Description

DESCRIPTION OF THE FIGURES

[0009] The following drawing figures, which form a part of this application, are illustrative of the described technology and are not meant to limit the scope of the disclosure in any manner.

[0010] FIG. 1 schematically illustrates an example of a flow cytometer system.

[0011] FIG. 2 schematically illustrates an example of a method of characterizing particles that can be performed by the flow cytometer system of FIG. 1.

[0012] FIG. 3 schematically illustrates an example of a method of measuring a first pulse width value of a particle that can be performed by the flow cytometer system of FIG. 1.

[0013] FIG. 4 graphically illustrates examples of first and second waveforms obtained in an operation of the method of FIG. 3.

[0014] FIG. 5 graphically illustrates examples of the first and second waveforms after completion of an operation in the method of FIG. 3.

[0015] FIG. 6 illustrates in more detail an example of an operation in the method of FIG. 3.

[0016] FIG. 7 graphically illustrates examples of the first and second waveforms after completion of the operation in FIG. 6.

[0017] FIG. 8 illustrates an example of another operation in the method of FIG. 3.

[0018] FIG. 9 graphically illustrates examples of the first and second waveforms after completion of a step in the operation of FIG. 8.

[0019] FIG. 10 graphically illustrates examples of first and second vectors generated for the first and second waveforms after completion of the operation of FIG. 8.

[0020] FIG. 11 graphically illustrates examples of measuring a second pulse width value of the first and second waveforms of FIG. 4.

[0021] FIG. 12 graphically illustrates data acquired from the flow cytometer system of FIG. 1 prior to characterizing the particles in accordance with the method of FIG. 2.

[0022] FIG. 13 graphically illustrates data acquired from the flow cytometer system of FIG. 1 after characterizing the particles in accordance with the method of FIG. 2, the particles being characterized using a first threshold.

[0023] FIG. 14 graphically illustrates data acquired from the flow cytometer system of FIG. 1 after characterizing the particles in accordance with the method of FIG. 2, the particles being characterized using a second threshold.

[0024] FIG. 15 graphically illustrates data acquired from the flow cytometer system of FIG. 1 after characterizing the particles in accordance with the method of FIG. 2, the particles being characterized using a third threshold.

[0025] FIG. 16 illustrates an exemplary architecture of a computing system that can be used to implement aspects of the flow cytometer system of FIG. 1.

DETAILED DESCRIPTION

[0026] Various embodiments will be described in detail with reference to the drawings, where like reference numerals represent like parts and assemblies throughout the several views. Reference to

various embodiments does not limit the scope of the claims attached hereto. Additionally, any examples set forth in this specification are not intended to be limiting and merely set forth some of the many possible embodiments for the appended claims.

[0027] FIG. 1 schematically illustrates an example of a flow cytometer system **100**. In general, flow cytometry is a technique for measuring and analyzing properties of particles or cells when flowing in a fluid stream. Data from millions of particles or cells can be collected by the flow cytometer system **100** in a matter of minutes and displayed in a variety of formats. Illustrative example applications of flow cytometry include phenotyping to identify and count specific cell types within a population, analyzing DNA or RNA content within cells, determining presence of antigens on a surface or within cells, and assessing cell health status.

[0028] As shown in the illustrative example of FIG. 1, the flow cytometer system **100** generally includes three main component subsystems: a fluidic system **110**, an optical system **120**, and an electronic system **130**. The fluidic system **110** includes a nozzle **112** which receives a sample containing particles or cells suspended in a fluid. The nozzle **112** creates a fluid stream **114** of the particles or cells arranged in a single file line. Each particle or cell passes through one or more light excitation beams produced by a light emitting unit **102**. The particles or cells intersect with the one or more light excitation beams at an interrogation zone **116**. In some examples, the light emitting unit **102** includes one or more light-emitting diodes (LEDs). In further examples, the light emitting unit **102** includes one or more lasers.

[0029] The optical system **120** includes the light emitting unit **102**, optical elements **122**, and detectors **124**. At the interrogation zone **116**, light from the light emitting unit **102** hits the particles or cells in the fluid stream **114** and scatters. The optical elements **122** direct the scattered light toward the detectors **124**. The detectors **124** can include a forward scatter (FSC) detector to measure scatter in the path of the light emitting unit **102**, a side scatter (SSC) detector to measure scatter at a ninety-degree angle relative to the light emitting unit **102**, one or more fluorescence detectors (FL1, FL2, FL3 . . . FLn) to measure the emitted fluorescence intensity at different wavelengths of light, and additional types of detectors.

[0030] Generally, FSC intensity is proportional to the size or diameter of a particle due to light diffraction around the particle. FSC may therefore be used for the discrimination of particles by size. SSC, on the other hand, is produced from light refracted or reflected by internal structures of the particle and may therefore provide information about the internal complexity or granularity of the particle. By adding fluorescent labelling to a sample, different fluorescent signals/channels (e.g., green, orange, and red) can be analyzed for functional characteristics of a cell. For example, since T-cells present CD3 binding sites, a sample containing T-cells may be “stained” with anti-CD3 antibodies conjugated with a fluorescent molecule. As these cells pass through the interrogation zone **116** in a single file line, the light from the source light excites the fluorescent tag, or fluorochrome, to emit photons at a wavelength detectable by a fluorescence detector. The detectors **124** may therefore simultaneously measure several parameters and enable categorization of particles by their function based on detected wavelengths of light.

[0031] The electronic system **130** includes a waveform acquisition device **140** and a waveform analysis device **150**. The waveform acquisition device **140** is communicatively coupled with the detectors **124** to receive analog waveform data **126** generated by the detectors **124**. The waveform acquisition device **140** includes an analog-to-digital converter (ADC) **142** that is configured to digitize the analog waveform data **126** received from the detectors **124**.

[0032] The waveform analysis device **150** is configured to receive the digital waveform data for processing and analysis. The waveform analysis device **150** can display the digital waveform data and analyses thereof on a graphical user interface (GUI) **152** for a user of the flow cytometer system **100**. In some examples, the waveform analysis device **150** includes a computing device **1600** communicatively coupled with a flow cytometer **101** such as through a wired or wireless connection. In some examples, the computing device **1600** can be communicatively coupled with

the flow cytometer **101** over a network such as a local area network (LAN) using Ethernet or Wi-Fi, or over a wide area network (WAN) such as the Internet. The flow cytometer **101** may include the fluidic system **110**, optical system **120**, and the waveform acquisition device **140**, but not the waveform analysis device **150**. In alternative embodiments, the waveform analysis device **150** is integrated with the flow cytometer **101**.

[0033] The flow cytometer system **100** includes elements which are shown and described for purposes of discussion, and it will be appreciated that numerous variations in components and functions are possible. The optical elements **122** may include a series of filters, dichroic mirrors, and/or beam splitters to select out different wavelengths of light and provide the wavelength to the appropriate detector. The detectors **124** may comprise, for example, photomultiplier tubes (PMTs) or avalanche photodiodes (APDs) or single photon counting devices.

[0034] FIG. 2 schematically illustrates an example of a method **200** of characterizing particles that can be performed by the flow cytometer system **100**. In some examples, the method **200** is performed by the waveform analysis device **150**.

[0035] The method **200** includes an operation **202** of determining a first pulse width value of a particle using a first technique. In some examples, the first technique includes directly measuring a pulse width of the particle. An example of the first technique is described by the method **300** shown in FIG. 3, which will be described in more detail below.

[0036] The method **200** includes an operation **204** of determining a second pulse width value of the particle using a second technique. In some examples, the second technique includes indirectly measuring the pulse width. An example of the second technique is shown in FIG. 9, which will be described in more detail below.

[0037] The method **200** includes an operation **206** of comparing the first and second pulse width values. In some examples, operation **206** includes determining a difference between the first and second pulse width values. In some examples, the difference between the first and second pulse width values is determined by Equation 1,

$$[00001] \text{ Difference} = C \cdot \text{Math.} \frac{W1 - W2}{W1 + W2} \cdot \text{Math.} \quad (1)$$

where W1 is the first pulse width value, and W2 is the second pulse width value. In view of Equation 1, the difference between the first and second pulse width values is determined by subtracting the second pulse width value from the first pulse width value, dividing by a sum of the first and second pulse width values, obtaining absolute value of result, and multiplying by a coefficient C. In some examples, the coefficient C is 2. The coefficient C may vary depending on the hardware of the flow cytometer system **100**. As described above, the first pulse width value W1 can include directly measuring the pulse width of the particle such as by performing the method **300** illustrated in FIG. 3, and the second pulse width value W2 can include indirectly measuring the pulse width of the particle.

[0038] The method **200** further includes an operation **208** of determining whether the difference between the first and second pulse width values exceeds a threshold. When the difference is equal to or less than the threshold (i.e., “No” in operation **208**), the method **200** proceeds to an operation **210** of characterizing the particle as a singlet particle. Otherwise, when the difference exceeds the threshold (i.e., “Yes” in operation **208**), the method **200** proceeds to an operation **212** of characterizing the particle as a concatenated particle. As used herein, a concatenated particle includes two or more particles that pass through the interrogation zone **116** at substantially the same time such as a doublet, triplet, quadruplet, and so on.

[0039] In some further examples, the method **200** can include characterizing pulse measurements of any shape including those distorted by saturation, noise, incorrect baseline correction, and the like from pulse measurements of a gaussian shape that are typical of single, isolated particles passing through the interrogation zone **116** (i.e., “singlets”). In some examples, the method **200** can further include sorting the particle into one or more groups based on the difference between the

first and second pulse width values. In such examples, the flow cytometer system **100** is a sorting flow cytometer that extracts particles out of the sample stream based on the difference between the first and second pulse width values calculated for the particles.

[0040] FIG. **3** schematically illustrates an example of a method **300** of determining a direct pulse width value of a particle. In some examples, the method **300** is performed as operation **202** of the method **200**. The method **300** can be performed by the flow cytometer system **100**. In some examples, the method is performed by the waveform analysis device **150**.

[0041] The method **300** includes an operation **302** of obtaining a waveform of a particle. For example, operation **302** can include the waveform analysis device **150** obtaining the waveform from the waveform acquisition device **140** after analog waveform data **126** received from the detectors **124** has been digitized by the ADC **142** of the waveform acquisition device **140**.

[0042] FIG. **4** graphically illustrates examples of first and second waveforms **12**, **14** obtained in operation **302** of the method **300**. In this example, the first and second waveforms **12**, **14** are from HEK293 cells analyzed by the flow cytometer system **100**. As described above, particles are constrained to pass in a single file line through the interrogation zone **116** where the particles intersect with the one or more light excitation beams from the light emitting unit **102**. This causes the particles to emit and/or scatter light, which is collected by the detectors **124** and converted into waveforms such as the first and second waveforms **12**, **14** shown in FIG. **4**. In this example, the first and second waveforms **12**, **14** are shown as each having experimental data points, and a solid line that is a gaussian fitting to the experimental data points.

[0043] In the example of FIG. **4**, the first waveform **12** has a gaussian shape having a single peak, which is representative of a single, isolated particle passing through the interrogation zone **116** (i.e., a “singlet”). The second waveform **14** has a non-gaussian shape having two peaks, which is representative of simultaneous illumination of two particles passing through the interrogation zone **116** (i.e., a “doublet”). Higher particle concentrations and/or increasing the cross-sectional dimension of the fluid stream **114** increases the probability of simultaneously illuminating two or more particles passing through the interrogation zone **116**. While the following disclosure refers to doublets, the techniques described herein can be similarly applied to detect other types of n-concatenated cells including triplets, quadruplets, and the like.

[0044] Referring back to FIG. **3**, the method **300** includes an operation **304** of setting a baseline to zero for the waveform obtained in operation **302**. Operation **304** can include determining a mean value for the baseline on both sides of the waveform of the particle, and subtracting the mean value from data points of the waveform to set the baseline to zero.

[0045] As shown in the example of FIG. **4**, the first waveform **12** has a baseline having a mean value of about 250 units, and operation **304** includes subtracting the mean value of about 250 units from all of the data points of the first waveform **12**. As a further example, the second waveform **14** has a baseline having a mean value of about—500 units, and operation **304** includes subtracting this mean value from all of the data points of the second waveform **14**. As used herein, the units represent voltage values from the detectors **124** that are digitized by the ADC **142** of the waveform acquisition device **140** using predefined sample rate and bit resolution.

[0046] FIG. **5** graphically illustrates examples of the first and second waveforms **12**, **14** after completion of the operation **304** in the method **300**. As shown in FIG. **5**, the first and second waveforms **12**, **14** each have their baseline set to zero. Also, as shown in FIGS. **4** and **5**, the maximum (M) value of the first waveform **12** decreased by about 250 units from about 3250 units in FIG. **4** to about 3000 units in FIG. **5**, the maximum (M) value of the second waveform **14** increased by about 500 units from about 4100 units in FIG. **4** to about 4600 units in FIG. **5**.

[0047] Referring back to FIG. **3**, the method **300** includes an operation **306** of folding the waveform in half. FIG. **6** schematically illustrates an example of operation **306** in more detail. Operation **306** can include a step **602** of determining the maximum (M) value of the waveform. In the example of FIG. **5**, the maximum (M) value of the first waveform **12** is about 3000 units, and

the maximum (M) value of the second waveform **14** is about 4600 units.

[0048] Operation **306** can further include a step **604** of calculating a half maximum (HM) value by dividing the maximum (M) value in two. In the illustrative example shown in FIG. 5, the half maximum (HM) value of the first waveform **12** is about **1500** units. As further shown in FIG. 5, and the half maximum (HM) value of the second waveform **14** is about **2300** units.

[0049] Operation **306** can further include a step **606** of subtracting the half maximum (HM) value from the data points of the waveform, and a step **608** of obtaining absolute values for the data points to generate a folded waveform. FIG. 7 graphically illustrates examples of the first and second waveforms **12**, **14** after completion of the operation **306** in the method **300**.

[0050] Referring back to FIG. 3, the method **300** includes an operation **308** of determining first and second vectors. FIG. 8 schematically illustrates an example of operation **308** in more detail.

Operation **308** can include a step **802** of determining a first minimum (l.sub.f) data point in the folded waveform, and a step **804** of determining a last minimum (l.sub.l) in the folded waveform.

[0051] In the example shown in FIG. 7, the first minimum (l.sub.f) data point of the first waveform **12** occurs at about 91 bin units, and the last minimum (l.sub.l) data point of the first waveform **12** occurs at about 110 bin units. The first minimum (l.sub.f) data point of the second waveform **14** occurs at about 70 bin units, and the last minimum (l.sub.l) data point of the second waveform **14** occurs at about 124 bin units. As used herein, 1 bin unit equals 0.15 (μ s).

[0052] Operation **308** can further include a step **806** of unfolding the waveform, a step **808** of selecting a first set of data points around the first minimum (l.sub.f) data point, and a step **810** of selecting a second set of data points around the last minimum (l.sub.l) data point. As an example, step **808** can include selecting about 5 data points around the first minimum (l.sub.f) data point, and step **810** can include selecting about 5 data points around the last minimum (l.sub.l) data point.

[0053] FIG. 9 graphically illustrates examples of the first and second waveforms **12**, **14** after completion of step **810** in operation **308**. As shown in FIG. 9, the first waveform **12** includes a first set of data points around the first minimum (lf) data point on a first side of the maximum (M) value of the first waveform **12**, and a second set of data points around the last minimum (l.sub.l) data point on a second side of the maximum (M) value of the first waveform **12**.

[0054] Similarly, as further shown in the example of FIG. 9, the second waveform **14** includes a first set of data points around the first minimum (lf) data point on a first side of the maximum (M) value of the second waveform **14**, and a second set of data points around the last minimum (l.sub.l) data point on a second side of the maximum (M) value of the second waveform **14**.

[0055] Operation **308** can further include a step **812** of generating the first vector by performing linear regression on the first set of data points, and a step **814** of generating the second vector by performing linear regression on the second set of data points. FIG. 10 graphically illustrates examples of the first and second vectors V1, V2 generated for the first and second waveforms **12**, **14** after completion of the operation **308** in the method **300**.

[0056] Referring back to FIG. 3, the method **300** includes an operation **310** of determining the first pulse width value as a distance between the first and second vectors V1, V2 at the half maximum value. In some instances, the first pulse width value is a full width at half maximum (FWHM). In this illustrative example, the first pulse width value of the first waveform **12** calculated in accordance with the method **300** is 19.00, and the first pulse width value calculated in accordance with the operations of the method **300** of the second waveform **14** is 54.38.

[0057] FIG. 11 graphically illustrates examples of measuring the second pulse width value of the first and second waveforms **12**, **14**. As shown in FIG. 11, measuring the second pulse width value can also include setting the baseline to zero (i.e., operation **304** of the method **300**) by determining a mean value for a baseline on both sides of a waveform of a particle, and subtracting the mean value from data points of the waveform.

[0058] The second pulse width value is an indirect measurement of the pulse width of the first and second waveforms **12**, **14** derived from the maximum (M) value and the pulse area (A). The second

pulse width value is calculated in accordance with Equation 2.

$$[00002] \text{PulseWidth} = 1.6651 \times (2) = 0.5642 \times \frac{A}{M} A = \text{sum}(Y)$$

[0059] In this illustrative example, the second pulse width value of the first waveform **12** is 19.14 bin units, and the second pulse width value of the second waveform **14** is 47.70 bin units. Table 1, as provided below, summarizes the first and second pulse width values of the first and second waveforms **12**, **14**, and the difference between these values calculated by Equation 1.

TABLE-US-00001

Waveform	First Pulse Width Value	Second Pulse Width Value	Difference
12	19.00	19.14	0.0073
14	54.38	47.70	0.1309

[0060] In view of the foregoing, the difference (0.0073) between the first and second pulse width values of the first waveform **12** is small, which is expected since the first waveform **12** has a gaussian shape that is typical of a single, isolated particle passing through the interrogation zone **116** (i.e., a “singlet”). In contrast, the difference (0.1309) between the first and second pulse width values of the second waveform **14** is large, which is expected since the second waveform **14** has a non-gaussian shape with two peaks, which is typical of simultaneous illumination of two particles passing through the interrogation zone **116** (i.e., a “doublet”). The calculated difference between the first and second pulse width values for a given waveform can be compared to a threshold for characterizing the waveform as a singlet or other type of particle.

[0061] FIG. **12** graphically illustrates data acquired from the flow cytometer system **100** prior to characterizing the particles in accordance with the method **200**. In the illustrative example of FIG. **12**, 3000 pulse measurements from a sample of HEK293 cells are obtained. A first plot **1202** includes pulse area (A) versus maximum (M) value. A second plot **1204** includes the second pulse width values determined in accordance with Equation 2. A third plot **1206** includes the first pulse width values determined in accordance with the method **300**.

[0062] FIG. **13** graphically illustrates data acquired from the flow cytometer system **100** after characterizing the particles in accordance with the method **200**. FIG. **13** includes first, second, and third plots **1302**, **1304**, **1306** of the same data shown in FIG. **12** of the 3000 pulse measurements from the sample of HEK293 cells. In FIG. **13**, the particles are characterized by using a first threshold of 0.1 such that the cells are characterized as concatenated cells (i.e., doublets, triplets, and the like) when the difference between the first and second pulse width values exceeds the first threshold. In this illustrative example, the first waveform **12** would be characterized as a singlet because the difference between the first and second pulse width values of the first waveform **12** is less than 0.1 (0.0073<0.1), and the second waveform **14** would be characterized as a doublet (or other concatenated particle) because the difference between the first and second pulse width values of the second waveform **14** is greater than 0.1 (0.1309>0.1).

[0063] FIG. **14** graphically illustrates data acquired from the flow cytometer system **100** after characterizing the particles in accordance with the method **200**. FIG. **14** includes first, second, and third plots **1402**, **1404**, **1406** of the same data shown in FIG. **12** of the 3000 pulse measurements from the sample of HEK293 cells. In FIG. **14**, the particles are characterized by using a second threshold of 0.07 such that the cells are characterized as concatenated cells (i.e., doublets, triplets, and the like) when the difference between the first and second pulse width values exceeds the second threshold. In this example, the first waveform **12** would be characterized as a singlet because the difference between the first and second pulse width values of the first waveform **12** is less than 0.07 (0.0073<0.07). The second waveform **14** would be characterized as a doublet (or other type of particle) because the difference between the first and second pulse width values of the second waveform **14** is greater than 0.07 (0.1309>0.07).

[0064] FIG. **15** graphically illustrates data acquired from the flow cytometer system **100** after characterizing the particles in accordance with the method **200**. FIG. **15** includes first, second, and third plots **1502**, **1504**, **1506** of the same data shown in FIG. **12** of the 3000 pulse measurements from the sample of HEK293 cells. In FIG. **15**, the particles are characterized by using a third

threshold of 0.03 such that the cells are characterized as concatenated cells (i.e., doublets, triplets, and the like) when the difference between the first and second pulse width values exceeds the third threshold. In this example, the first waveform **12** would be characterized as a singlet because the difference between the first and second pulse width values of the first waveform **12** is less than 0.03 ($0.0073 < 0.03$). The second waveform **14** would be characterized as a doublet (or other type of particle) because the difference between the first and second pulse width values of the second waveform **14** is greater than 0.03 ($0.1309 > 0.03$).

[0065] Given the foregoing, the method **200** is an objective approach to distinguishing single, isolated particles passing through the interrogation zone **116** from other types of particles such as concatenated particles (e.g., doublets, triplets, and the like), also from pulse measurements distorted by saturation, noise, wrong baseline correction, and the like. Furthermore, the threshold value (e.g., first threshold of 0.1, second threshold of 0.07, and third threshold of 0.03) can be adjusted as desired to achieve different level(s) of pulse shape discrimination. Also, the pulse shape discrimination by the method **200** is automatic such that it does not require prior populations to set gates for further discrimination, and can instead be performed on the fly. This can result in higher yields. Also, the pulse shape discrimination by the method **200** can be performed on a field-programmable gate array (FPGA) that is an integrated circuit designed to be configured by a customer or a designer after manufacturing.

[0066] FIG. **16** schematically illustrates an exemplary architecture of a computing device **1600** for implementing aspects of the flow cytometer system **100** including aspects performed by the waveform analysis device **150**. The computing device **1600** includes one or more processing devices **1602**, a memory storage device **1604**, and a system bus **1606** coupling the memory storage device **1604** to the one or more processing devices **1602**.

[0067] The one or more processing devices **1602** can include a processor such as a central processing unit (CPU). The one or more processing devices **1602** can include a microcontroller having one or more digital signal processors, field-programmable gate arrays (FPGA), and/or other types of electronic circuits. In some further examples, the one or more processing devices **1602** are part of a processing circuitry having non-transitory computer readable storage media storing instructions which, when executed by the processing circuitry, cause the processing circuitry to perform the aspects and functionalities described herein.

[0068] The memory storage device **1604** can include a random-access memory ("RAM") **1608** and a read-only memory ("ROM") **1610**. Basic input and output logic having basic routines transferring information between elements in the flow cytometer system **100** can be stored in the ROM **1610**. The flow cytometer system **100** can additionally include a mass storage device **1612** that can store an operating system **1614** and software instructions **1616**. The mass storage device **1612** is connected to the one or more processing devices **1602** through the system bus **1606**. The mass storage device **1612** and computer-readable data storage media provide non-volatile, non-transitory computer memory storage.

[0069] Although the description of computer-readable data storage media contained herein refers to the mass storage device **1612**, it should be appreciated by those skilled in the art that computer-readable data storage media can be any available non-transitory, physical device or article of manufacture from which the flow cytometer system **100** can read data and/or instructions. The computer-readable storage media can be comprised of entirely non-transitory media. The mass storage device **1612** is an example of a computer-readable storage device.

[0070] Computer-readable data storage media include volatile and non-volatile, removable, and non-removable, media implemented in any method or technology for storage of information such as computer-readable software instructions, data structures, program modules or other data. Example types of computer-readable data storage media include, but are not limited to, RAM, ROM, EPROM, EEPROM, flash memory or other solid-state memory technology, or any other medium which can be used to store information, and which can be accessed by the device.

[0071] The flow cytometer system **100** can operate in a networked environment using logical connections to the other devices through a communications network **1620**. For example, the computing device **1600** connects to the communications network **1620** through a network interface unit **1618** connected to the system bus **1606**. The network interface unit **1618** can connect to other types of communications networks and devices, including through Bluetooth, Wi-Fi, Ethernet, and cellular telecommunications networks. The network interface unit **1618** can connect the computing device **1600** to additional networks, systems, and devices. The computing device **1600** also includes an input/output unit **1622** for receiving and processing inputs and outputs from one or more peripheral devices, and the user interface **152**.

[0072] The mass storage device **1612** and the RAM **1608** store software instructions and data. The software instructions can include an operating system **1614** suitable for controlling the operation of the computing device **1600**. The mass storage device **1612** and/or the RAM **1608** can also store the software instructions **1616**, which when executed by the one or more processing devices **1602**, provide the functionalities and aspects described herein.

[0073] The various embodiments described above are provided by way of illustration only and should not be construed to be limiting in any way. Various modifications can be made to the embodiments described above without departing from the true spirit and scope of the disclosure.

[0074] Embodiments of the disclosure can be described with reference to the following numbered clauses, with preferred features laid out in the dependent clauses:

[0075] 1. A method of characterizing particles in flow cytometry, the method comprising: [0076] determining a first pulse width value of a particle using a first technique; [0077] determining a second pulse width value of the particle using a second technique; [0078] comparing the first and second pulse width values; and [0079] characterizing the particle as a concatenated particle when a difference between the first and second pulse width values exceeds a threshold.

[0080] 2. The method of clause 1, wherein the first technique includes directly measuring a pulse width of the particle, and the second technique includes indirectly measuring the pulse width.

[0081] 3. The method of clause 2, wherein the first and second techniques each include: [0082] determining a mean value for a baseline on both sides of a waveform of the particle; and [0083] subtracting the mean value from data points of the waveform to set the baseline to zero.

[0084] 4. The method of clause 3, wherein the first technique further includes: [0085] determining a maximum value of the waveform; [0086] calculating a half maximum value by dividing the maximum value by two; [0087] subtracting the half maximum value from the data points of the waveform; and [0088] obtaining absolute values for the data points to generate a folded waveform.

[0089] 5. The method of clause 4, wherein the first technique further includes: [0090] determining a first minimum and a last minimum in the folded waveform; [0091] selecting a first set of data points around the first minimum; and [0092] selecting a second set of data points around the last minimum.

[0093] 6. The method of clause 5, wherein the first technique further includes: [0094] generating a first vector by performing linear regression on the first set of data points; [0095] generating a second vector by performing linear regression on the second set of data points; and [0096] determining the first pulse width value as a distance between the first and second vectors at the half maximum value.

[0097] 7. The method as in any one of the preceding clauses, wherein the second technique further includes: [0098] measuring the second pulse width value as a function of the maximum value of the waveform and an area of the waveform.

[0099] 8. The method as in any one of the preceding clauses, wherein the difference between the first and second pulse width values is calculated by: [0100] subtracting the second pulse width value from the first pulse width value; [0101] dividing by a sum of the first and second pulse width values; [0102] obtaining an absolute value; and [0103] multiplying the absolute value by a coefficient.

[0104] 9. The method as in any one of the preceding clauses, wherein the threshold is 0.1.

[0105] 10. The method as in any one of the preceding clauses, wherein the threshold is 0.07.

[0106] 11. The method as in any one of the preceding clauses, wherein the threshold is 0.03.

[0107] 12. The method as in any one of the preceding clauses, further comprising: [0108] characterizing the particle as a singlet particle when the difference between the first and second pulse width values is equal to or less than the threshold.

[0109] 13. The method as in any one of the preceding clauses, further comprising: [0110] characterizing the particle as having a gaussian shape when the difference between the first and second pulse width values is equal to or less than the threshold.

[0111] 14. The method as in any one of the preceding clauses, further comprising: [0112] characterizing the particle as having a non-gaussian shape when the difference between the first and second pulse width values exceeds the threshold.

[0113] 15. The method as in any one of the preceding clauses, further comprising: [0114] sorting the particle into one or more groups based on the difference between the first and second pulse width values.

[0115] 16. A system for characterizing particles, the system comprising: [0116] a light emitting unit generating one or more excitation light beams; [0117] a fluidic system streaming the particles through the one or more excitation light beams at an interrogation zone; [0118] optical elements directing scattered light from the interrogation zone; [0119] one or more detectors measuring characteristics of the scattered light; and [0120] a processing circuitry having non-transitory computer readable storage media storing instructions which, when executed by the processing circuitry, cause the processing circuitry to: [0121] determine first pulse width values of the particles using a first technique; [0122] determine second pulse width values of the particles using a second technique; [0123] compare the first and second pulse width values; and [0124] characterize the particles as concatenated particles when a difference between the first and second pulse width values exceeds a threshold.

[0125] 17. The system of clause 16, wherein the first technique includes directly measuring a pulse width of the particles, and the second technique includes indirectly measuring the pulse width.

[0126] 18. The system of clause 17, wherein the first and second techniques each include: [0127] determine a mean value for a baseline on both sides of a waveform of a particle; and [0128] subtract the mean value from data points of the waveform to set the baseline to zero.

[0129] 19. The system of clause 18, wherein the first technique further includes: [0130] determine a maximum value of the waveform; [0131] calculate a half maximum value by dividing the maximum value by two; [0132] subtract the half maximum value from the data points of the waveform; and [0133] obtain absolute values for the data points to generate a folded waveform.

[0134] 20. The system of clause 19, wherein the first technique further includes: [0135] determine a first minimum and a last minimum in the folded waveform; [0136] select a first set of data points around the first minimum; and [0137] select a second set of data points around the last minimum.

[0138] 21. The system of clause 20, wherein the first technique further includes: [0139] generate a first vector by performing linear regression on the first set of data points; [0140] generate a second vector by performing linear regression on the second set of data points; and [0141] measure the first pulse width value as a distance between the first and second vectors at the half maximum value.

[0142] 22. The system as in any of clauses 16-21, wherein the second technique further includes: [0143] measure the second pulse width value as a function of the maximum value of the waveform and an area of the waveform.

[0144] 23. The system as in any of clauses 16-22, wherein the difference between the first and second pulse width values is calculated by: [0145] subtracting the second pulse width value from the first pulse width value; [0146] dividing by a sum of the first and second pulse width values; [0147] obtaining an absolute value; and [0148] multiplying the absolute value by a coefficient.

[0149] 24. The system as in any of clauses 16-23, wherein the threshold is 0.1.
[0150] 25. The system as in any of clauses 16-23, wherein the threshold is 0.07.
[0151] 26. The system as in any of clauses 16-23, wherein the threshold is 0.03.
[0152] 27. A method of characterizing a particle in flow cytometry, the method comprising:
obtaining a waveform of the particle; [0153] determining a mean value of a baseline on both sides
of the waveform of the particle; [0154] subtracting the mean value from data points of the
waveform; [0155] determining a maximum value of the waveform; [0156] calculating a half
maximum value by dividing the maximum value by two; [0157] subtracting the half maximum
value from the data points of the waveform; [0158] obtaining absolute values for the data points to
generate a folded waveform; [0159] determining a first minimum and a last minimum in the folded
waveform; [0160] generating a first vector by performing linear regression on a first set of data
points in proximity to the first minimum; [0161] generating a second vector by performing linear
regression on a second set of data points in proximity to the last minimum; and [0162] measuring a
direct pulse width value of the particle as a distance between the first and second vectors at the half
maximum value.
[0163] 28. The method of clause 27, further comprising: [0164] measuring an indirect pulse width
value as a function of the maximum value of the waveform and an area of the waveform.
[0165] 29. The method of clause 28, further comprising: [0166] characterizing the particle as a
concatenated particle when a difference between the direct pulse width value and the indirect pulse
width value exceeds a threshold.
[0167] 30. The method of clause 29, further comprising: [0168] calculating the difference between
the direct pulse width value and the indirect pulse width value by: [0169] subtracting the indirect
pulse width value from the direct pulse width value; [0170] dividing by a sum of the direct and
indirect pulse width values; [0171] obtaining an absolute value; and [0172] multiplying the
absolute value by a coefficient.
[0173] 31. The method of clauses 29 or 30, wherein the threshold is 0.1.
[0174] 32. The method of clauses 29 or 30, wherein the threshold is 0.07.
[0175] 33. The method of clauses 29 or 30, wherein the threshold is 0.03.

Claims

1. A system for characterizing particles, the system comprising: a light emitting unit generating one or more excitation light beams; a fluidic system streaming the particles through the one or more excitation light beams at an interrogation zone; optical elements directing scattered light from the interrogation zone; one or more detectors measuring characteristics of the scattered light; and a processing circuitry having non-transitory computer readable storage media storing instructions which, when executed by the processing circuitry, cause the processing circuitry to: determine first pulse width values of the particles using a first technique; determine second pulse width values of the particles using a second technique; compare the first and second pulse width values; and characterize the particles as concatenated particles when a difference between the first and second pulse width values exceeds a threshold.
2. The system of claim 1, wherein the first technique includes directly measuring a pulse width of the particles, and the second technique includes indirectly measuring the pulse width.
3. The system of claim 2, wherein the first and second techniques each include: determine a mean value for a baseline on both sides of a waveform of a particle; and subtract the mean value from data points of the waveform to set the baseline to zero.
4. The system of claim 3, wherein the first technique further includes: determine a maximum value of the waveform; calculate a half maximum value by dividing the maximum value by two; subtract the half maximum value from the data points of the waveform; and obtain absolute values for the data points to generate a folded waveform.

5. The system of claim 4, wherein the first technique further includes: determine a first minimum and a last minimum in the folded waveform; select a first set of data points around the first minimum; and select a second set of data points around the last minimum.
 6. The system of claim 5, wherein the first technique further includes: generate a first vector by performing linear regression on the first set of data points; generate a second vector by performing linear regression on the second set of data points; and measure the first pulse width value as a distance between the first and second vectors at the half maximum value.
 7. The system of claim 1, wherein the second technique further includes: measure the second pulse width value as a function of the maximum value of the waveform and an area of the waveform.
 8. The system of claim 1, wherein the difference between the first and second pulse width values is calculated by: subtracting the second pulse width value from the first pulse width value; dividing by a sum of the first and second pulse width values; obtaining an absolute value; and multiplying the absolute value by a coefficient.
 9. The system of claim 1, wherein the threshold is 0.1.
 10. The system of claim 1, wherein the threshold is 0.07.
 11. The system of claim 1, wherein the threshold is 0.03.
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